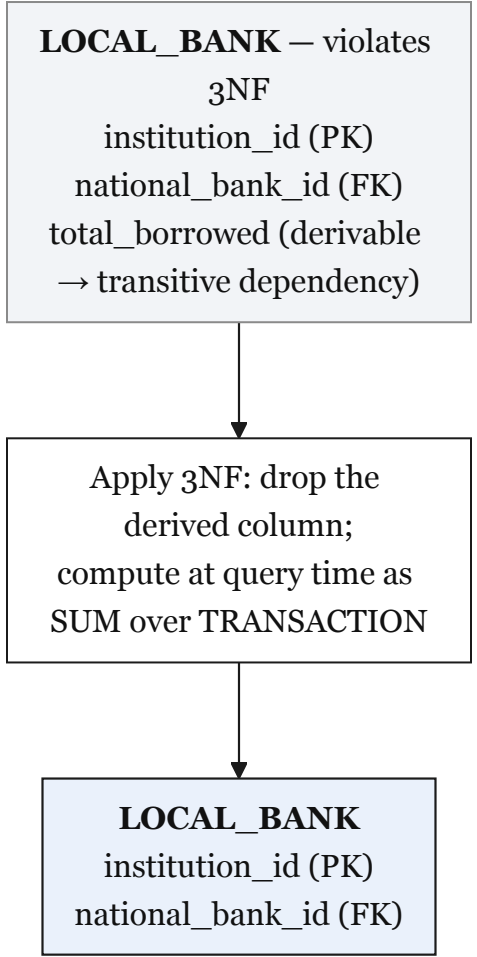


LOCAL_BANK — violates
3NF
institution_id (PK)
national_bank_id (FK)
total_borrowed (derivable
→ transitive dependency)



Apply 3NF: drop the
derived column;
compute at query time as
SUM over TRANSACTION

LOCAL_BANK
institution_id (PK)
national_bank_id (FK)